



11.0.1 UGC NET CSE | June 2013 | Part 2 | Question: 39



Which of the following shall be a compound proposition involving the propositions p , q and r , that is true when exactly two of the p , q and r are true and is false otherwise?

- A. $(p \vee q \wedge r) \vee (p \wedge q \wedge r) \wedge (\neg p \wedge q \vee r)$
 B. $(p \wedge q \vee r) \wedge (p \wedge q \wedge r) \vee (\neg q \wedge q \wedge r)$
 C. $(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$
 D. $(p \vee r \wedge q) \vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$

ugcnetcse-june2013-paper2

Answer key

11.0.2 UGC NET CSE | June 2013 | Part 2 | Question: 40



The truth value of the statements:

$\exists! x P(x) \rightarrow \exists x P(x)$ and $\exists! x \neg P(x) \rightarrow \neg \forall x P(x)$, (where the notation $\exists! x P(x)$ denotes the proposition “There exists a unique x such that $P(x)$ is true”) are:

- A. True and False B. False and True C. False and False D. True and True

ugcnetcse-june2013-paper2

Answer key

11.0.3 UGC NET CSE | September 2013 | Part 3 | Question: 67



The tracing model in Prolog describes program execution in terms of certain events. These events are

- A. call and exit B. call and fail
 C. call, exit and redo D. call, exit, redo and fail

ugcnetcse-sep2013-paper3

11.0.4 UGC NET CSE | December 2015 | Part 2 | Question: 6



Which of the following arguments are not valid?

- i. "If Gora gets the job and works hard, then he will be promoted. if Gora gets promotion, then he will be happy. He will not be happy, therefore, either he will not get the job or he will not work hard."
 ii. "Either Puneet is not guilty or Pankaj is telling the truth. Pankaj is not telling the truth, therefore, Puneet is not guilty."
 iii. If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$, then $n > 1$.

- A. i and iii B. ii and iii C. i, ii, and iii D. i and ii

ugcnetcse-dec2015-paper2 discrete-mathematics mathematical-logic

Answer key

11.0.5 UGC NET CSE | June 2014 | Part 3 | Question: 32



_____ predicate calculus allows quantified variables to refer to objects in the domain of discourse and not to predicates or functions.

- A. Zero-order B. First-order
 C. Second-order D. High-order

ugcnetjune2014iii discrete-mathematics mathematical-logic

Answer key

11.0.6 UGC NET CSE | Junet 2015 | Part 2 | Question: 8

"If my computations are correct and I pay the electric bill, then I will run out of money. If I don't pay the electric bill, the power will be turned off. Therefore, If I don't run out of money and the power is still on then my computations are incorrect."

Convert this argument into logical notations using the variables c, b, r, p for propositions of computations, electric bills, out of money and the power respectively. (Where $\neg$ means NOT).

- A. if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$
- B. if $(c \vee b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(r \wedge p) \rightarrow c$
- C. if $(c \wedge b) \rightarrow r$ and $\neg p \rightarrow \neg b$, then $(\neg r \vee p) \rightarrow \neg c$
- D. if $(c \vee b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$

ugcnetcse-june2015-paper2 mathematical-logic

Answer key

11.0.7 UGC NET CSE | December 2014 | Part 3 | Question: 55

Equivalent logical expression for the Well Formed Formula (WFF),

$$\sim (\forall x)F[x]$$

is

- A. $\forall x(\sim F[x])$
- B. $\sim (\exists x)F[x]$
- C. $\exists x(\sim F[x])$
- D. $\forall xF[x]$

ugcnetcse-dec2014-paper3 mathematical-logic

Answer key

11.0.8 UGC NET CSE | December 2014 | Part 3 | Question: 57

The resolvent of the set of clauses $(A \vee B, \sim A \vee D, C \vee \sim B)$ is

- A. $A \vee B$
- B. $C \vee D$
- C. $A \vee C$
- D. $A \vee D$

ugcnetcse-dec2014-paper3 mathematical-logic

Answer key

11.0.9 UGC NET CSE | Junet 2015 | Part 3 | Question: 22

The casual form of the disjunctive normal form $\neg A \vee \neg B \vee \neg C \vee D$ is:

- A. $A \wedge B \wedge C \Rightarrow D$
- B. $A \vee B \vee C \vee D \Rightarrow \text{true}$
- C. $A \wedge B \wedge C \wedge D \Rightarrow \text{true}$
- D. $A \wedge B \wedge C \wedge D \Rightarrow \text{false}$

ugcnetcse-june2015-paper3 mathematical-logic

Answer key

11.0.10 UGC NET CSE | December 2015 | Part 2 | Question: 7

Let $P(m, n)$ be the statement "m divides n" where the universe of discourse for both the variable is the set of positive integers. Determine the truth values of each of the following propositions:

- i. $\forall m \forall n P(m, n)$,
- ii. $\forall n P(1, n)$
- iii. $\exists m \forall n P(m, n)$
 - A. a-True, b-True, c-False
 - B. a-True, b-False, c-False
 - C. a-False, b-False, c-False

D. a-True, b-True, c-True

ugcnetcse-dec2015-paper2 discrete-mathematics mathematical-logic

Answer key 

11.0.11 UGC NET CSE | December 2015 | Part 2 | Question: 9



Consider the compound propositions given below as:

- i. $p \vee \sim (p \wedge q)$
- ii. $(p \wedge \sim q) \vee \sim (p \wedge q)$
- iii. $p \wedge (q \vee r)$

Which of the above propositions are tautologies

- A. i and iii B. ii and iii C. i and ii D. i, ii, and iii

ugcnetcse-dec2015-paper2 mathematical-logic

Answer key 

11.0.12 UGC NET CSE | June 2011 | Part 2 | Question: 3



The proposition $\sim p \vee q$ is equivalent to

- A. $p \rightarrow q$ B. $q \rightarrow p$
C. $p \leftrightarrow q$ D. $p \vee q$

ugcnetcse-june2011-paper2 discrete-mathematics mathematical-logic

Answer key 

11.0.13 UGC NET CSE | November 2017 | Part 2 | Question: 8



Let P and Q be two propositions $\neg(P \leftrightarrow Q)$ is equivalent to

- A. $P \leftrightarrow \neg Q$ B. $\neg P \leftrightarrow Q$
C. $\neg P \leftrightarrow \neg Q$ D. $Q \rightarrow P$

ugcnetcse-nov2017-paper2 discrete-mathematics mathematical-logic

Answer key 

11.0.14 UGC NET CSE | December 2018 | Part 2 | Question: 2



Match List-I with List-II and choose the correct answer from the code given below :

	List I		List II
(a)	Equivalence	(i)	$p \Rightarrow q$
(b)	Contrapositive	(ii)	$p \Rightarrow q; q \Rightarrow p$
(c)	Converse	(iii)	$p \Rightarrow q : \sim q \Rightarrow \sim p$
(d)	Implication	(iv)	$p \Leftrightarrow q$

- A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) B. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
C. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

ugcnetcse-dec2018-paper2 mathematical-logic

Answer key 

11.0.15 UGC NET CSE | December 2018 | Part 2 | Question: 1



In mathematical logic, which of the following are statements?

- i. There will be snow in January.
- ii. What is the time now?
- iii. Today is Sunday.

iv. You must study Discrete mathematics

Choose the correct answer from the code given below:

- A. i and iii B. i and ii C. ii and iv D. iii and iv

ugcnetcse-dec2018-paper2 mathematical-logic

Answer key 

11.0.16 UGC NET CSE | January 2017 | Part 2 | Question: 2



Match the following :

List – I

List – II

- | | |
|---------------|---|
| a. Absurd | i. Clearly impossible being contrary to some evident truth. |
| b. Ambiguous | ii. Capable of more than one interpretation or meaning. |
| c. Axiom | iii. An assertion that is accepted and used without a proof. |
| d. Conjecture | iv. An opinion preferably based on some experience or wisdom. |

Codes:

- | | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> |
|----|-----------|------------|------------|-----------|
| A. | <i>i</i> | <i>ii</i> | <i>iii</i> | <i>iv</i> |
| B. | <i>i</i> | <i>iii</i> | <i>iv</i> | <i>ii</i> |
| C. | <i>ii</i> | <i>iii</i> | <i>iv</i> | <i>i</i> |
| D. | <i>ii</i> | <i>i</i> | <i>iii</i> | <i>iv</i> |

ugcnetjan2017ii mathematical-logic

Answer key 

11.0.17 UGC NET CSE | October 2020 | Part 2 | Question: 53



Consider the following statements:

- i. Any tree is 2-colorable
- ii. A graph G has no cycles of even length if it is bipartite
- iii. A graph G is 2-colorable if it is bipartite
- iv. A graph G can be colored with $d + 1$ colors if d is the maximum degree of any vertex in the graph G
- v. A graph G can be colored with $O(\log |v|)$ colors if it has $O(|v|)$ edges.

Choose the correct answer from the options given below:

- A. (iii) and (v) are incorrect B. (ii) and (iii) are incorrect
C. (ii) and (v) are incorrect D. (i) and (iv) are incorrect

ugcnetcse-oct2020-paper2 discrete-mathematics graph-theory

Answer key 

11.0.18 Computer Science - UGC NET 2021 | Question ID = 2352 |



Which of these statements about the floor and ceiling functions are correct ?

Statement 1 : $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + (1/2) \rfloor$ for all real numbers x .

Statement 2 : $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$ for all real numbers x and y .

- 1. Both Statement I and Statement II are true.
- 2. Both Statement I and Statement II are false.
- 3. Statement I is true but Statement II is false.

4. Statement I is false but Statement II is true.

discrete-mathematics mathematical-logic engineering-mathematics

Answer key 

11.0.19 UGC NET CSE_DEC 2023



If universe of discourse are all real numbers, then which of the following are true?

- (A) $\exists x \forall y (x + y = y)$
- (B) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
- (C) $\exists x \exists y (((x \leq 0) \wedge (y \leq 0)) \wedge (x - y > 0))$
- (D) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

Choose the **correct** answer from the options given below:

- (1) (A) and (B) Only
- (2) (A), (C) and (D) Only
- (3) (A), (B) and (D) Only
- (4) (A), (B), (C) and (D) Only

Answer key 

11.1

Countable Uncountable Set (1)

11.1.1 Countable Uncountable Set: UGC NET CSE | December 2015 | Part 2 | Question: 3



Which of the following is/are not true ?

- i. The set of negative integers is countable.
- ii. The set of integers that are multiples of 7 is countable.
- iii. The set of even integers is countable.
- iv. The set of real numbers between 0 and $1/2$ is countable.

- A. i and iii B. ii and iv C. ii only D. iv only

ugcnetcse-dec2015-paper2 discrete-mathematics set-theory countable-uncountable-set

Answer key 

11.2

First Order Logic (5)

11.2.1 First Order Logic: UGC NET CSE | January 2017 | Part 3 | Question: 60



The first order logic (FOL) statement $((R \vee Q) \wedge (P \vee \neg Q))$ is equivalent to which of the following?

- A. $((R \vee \neg Q) \wedge (P \vee \neg Q) \wedge (R \vee P))$
- B. $((R \vee Q) \wedge (P \vee \neg Q) \wedge (R \vee P))$
- C. $((R \vee Q) \wedge (P \vee \neg Q) \wedge (R \vee \neg P))$
- D. $((R \vee Q) \wedge (P \vee \neg Q) \wedge (\neg R \vee P))$

ugcnetcse-jan2017-paper3 mathematical-logic first-order-logic

Answer key 

11.2.2 First Order Logic: UGC NET CSE | July 2018 | Part 2 | Question: 77



Consider the following English sentence:

"Agra and Gwalior are both in India".

A student has written a logical sentence for the above English sentence in First-Order Logic using predicate $IN(x, y)$, which means x is in y , as follows.

$\text{In(Agra, India)} \vee \text{In(Gwalior, India)}$

Which one of the following is correct with respect to the above logical sentence?

- A. It is syntactically valid but does not express the meaning of the English sentence
- B. It is syntactically valid and expresses the meaning of the English sentence also
- C. It is syntactically invalid but expresses the meaning of the English sentence
- D. It is syntactically invalid and does not express the meaning of the English sentence

ugcnetcse-july2018-paper2 discrete-mathematics first-order-logic

Answer key 

11.2.3 First Order Logic: UGC NET CSE | June 2013 | Part 3 | Question: 69



If we convert

$\exists u \forall v \forall x \exists y (P(f(u), v, x, y) \rightarrow Q(u, v, y))$ to $\forall v \forall x (P(f(a), v, x, g(v, x)) \rightarrow Q(a, v, g(v, x)))$.

This process is known as

- A. Simplification
- B. Unification
- C. Skolemization
- D. Resolution

ugcnetcse-june2013-paper3 discrete-mathematics first-order-logic

Answer key 

11.2.4 First Order Logic: UGC NET CSE | June 2014 | Part 2 | Question: 19



The notation $\exists! x P(x)$ denotes the proposition “there exists a unique x such that $P(x)$ is true”. Give the truth values of the following statements :

I. $\exists! x P(x) \rightarrow \exists x P(x)$

II. $\exists! x \neg P(x) \rightarrow \neg \forall x P(x)$

- A. Both I & II are true.
- B. Both I & II are false.
- C. I - false, II - true
- D. I - true, II - false

ugcnetcse-june2014-paper2 mathematical-logic first-order-logic

Answer key 

11.2.5 First Order Logic: UGC NET CSE | September 2013 | Part 3 | Question: 1



Which of the following is a correct predicate logic statement for “Every Natural number has one successor”?

- A. $\forall x \exists y (succ(x, y) \wedge (\exists z succ(x, z) \Rightarrow equal(y, z)))$
- B. $\forall x \exists y (succ(x, y) \vee (\exists z succ(x, z) \Rightarrow equal(y, z)))$
- C. $\exists x \forall y (succ(x, y) \wedge (\exists z succ(x, z) \Rightarrow equal(y, z)))$
- D. $\forall x \exists y (succ(x, y))$

first-order-logic ugcnetcse-sep2013-paper3

Answer key 

11.3 Functions (1)

11.3.1 Functions: UGC NET CSE | July 2018 | Part 2 | Question: 87



Match the following in **List-I** and **List-II**, for a function f :

List-I

- (a) $\forall x \forall y (f(x) = f(y) \rightarrow x = y)$
 (b) $\forall y \exists x (f(x) = y)$
 (c) $\forall x f(x) = k$

List-II

- (i) Constant
 (ii) Injective
 (iii) Surjective

Code :

- A. (a)-(i), (b)-(ii), (c)-(iii)
 C. (a)-(ii), (b)-(i), (c)-(iii)

- B. (a)-(iii), (b)-(ii), (c)-(i)
 D. (a)-(ii), (b)-(iii), (c)-(i)

ugcnetcse-july2018-paper2 discrete-mathematics functions

Answer key **11.4****Group Theory (1)****11.4.1 Group Theory: UGC NET CSE | December 2015 | Part 2 | Question: 10**

Which of the following property/ies a Group G must hold, in order to be an Abelian group?

- i. The distributive property
 ii. The commutative property
 iii. The symmetric property

- A. i and ii B. ii and iii C. i only D. ii only

ugcnetcse-dec2015-paper2 discrete-mathematics set-theory&algebra group-theory

Answer key **11.5****Prolog (1)****11.5.1 Prolog: UGC NET CSE | December 2015 | Part 3 | Question: 7**

Given the following set of prolog clauses:

```
father(X,Y) :
parent(X,Y),
male(X),
parent(Sally, Bob),
parent(Jim, Bob),
parent(Alice, Jane),
male(Bob),
male(Jim),
female(Salley),
female(Alice)
```

How many atoms are matched to the variable 'X' before the query father(X, Jane) reports a Result?

- A. 1 B. 2 C. 3 D. 4

ugcnetcse-dec2015-paper3 mathematical-logic prolog non-gatecse

Answer key **11.6****Propositional Logic (15)****11.6.1 Propositional Logic: Computer Science - UGC NET 2021 | Question ID = 2351 |**

Which of the following are logically equivalent ?

- i. $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$
 ii. $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$
 iii. $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$

Choose the correct answer from the options given :

- A. i and ii only B. i and iii only
 C. i only D. ii and iii only

Answer key **11.6.2 Propositional Logic: UGC NET CSE | December 2006 | Part 2 | Question: 2**

The proposition $\sim q \vee p$ is equivalent to :

- A. B. C. D.

2. The proposition $\sim q \vee p$ is equivalent to :

- (A) (B) (C) (D)

ugcnetcse-dec2006-paper2 mathematical-logic propositional-logic

Answer key **11.6.3 Propositional Logic: UGC NET CSE | December 2013 | Part 2 | Question: 34**

Let $P(m,n)$ be the statement "m divides n" where the universe of discourse for both the variable is the set of positive integers. Determine the truth values of each of the following propositions:

I. $\forall m \forall n P(m,n)$,

II. $\exists m \forall n P(m,n)$

A. Both I and II are true

C. I-false & II-true

B. Both I and II are false

D. I-true& II-false

ugcnetcse-dec2013-paper2 discrete-mathematics propositional-logic

Answer key **11.6.4 Propositional Logic: UGC NET CSE | December 2015 | Part 3 | Question: 44**

In propositional logic, given P and $P \rightarrow Q$, we can infer _____

A. $\sim Q$

C. $P \wedge Q$

B. Q

D. $\sim P \wedge Q$

ugcnetcse-dec2015-paper3 propositional-logic mathematical-logic

Answer key **11.6.5 Propositional Logic: UGC NET CSE | January 2017 | Part 2 | Question: 6**

In propositional logic if $(P \rightarrow Q) \wedge (R \rightarrow S)$ and $(P \vee R)$ are two premises such that

$$\frac{\frac{(P \rightarrow Q) \wedge (R \rightarrow S)}{P \vee R}}{Y}$$

Y is the premise :

A. $P \vee R$

B. $P \vee S$

C. $Q \vee R$

D. $Q \vee S$

ugcnetjan2017ii discrete-mathematics propositional-logic

Answer key **11.6.6 Propositional Logic: UGC NET CSE | January 2017 | Part 3 | Question: 59**

Which of the following statements is true?

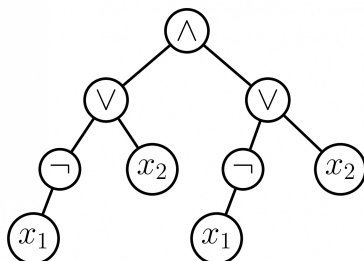
- A. The sentence S is a logical consequence of $S_1, \dots, S_n$ if and only if $S_1 \wedge S_2 \wedge \dots \wedge S_n \rightarrow S$ is satisfiable.
 B. The sentence S is a logical consequence of $S_1, \dots, S_n$ if and only if $S_1 \wedge S_2 \wedge \dots \wedge S_n \rightarrow S$ is valid.
 C. The sentence S is a logical consequence of $S_1, \dots, S_n$ if and only if $S_1 \wedge S_2 \wedge \dots \wedge S_n \rightarrow S$ is consistent.
 D. The sentence S is a logical consequence of $S_1, \dots, S_n$ if and only if $S_1 \wedge S_2 \wedge \dots \wedge S_n \wedge S$ is inconsistent.

Answer key

11.6.7 Propositional Logic: UGC NET CSE | June 2009 | Part 2 | Question: 25



The propositional formula given by the tree :
is :



- A. $x_2 \wedge x_1 \vee \neg x_1 \vee \neg x_1$
 B. $(x_2 \vee \neg x_2) \wedge (x_1 \vee x_2)$
 C. $(\neg x_1 \vee x_2) \wedge (\neg x_1 \vee x_2)$
 D. None

Answer key

11.6.8 Propositional Logic: UGC NET CSE | June 2014 | Part 2 | Question: 20



Give a compound proposition involving propositions p, q and r that is true when exactly two of p, q and r are true and is false otherwise

- A. $(p \vee q \wedge \neg r) \wedge (p \wedge \neg q \wedge r) \wedge (\neg p \wedge q \wedge r)$
 B. $(p \wedge q \wedge \neg r) \wedge (p \vee q \wedge \neg r) \wedge (\neg p \wedge q \wedge r)$
 C. $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \wedge (\neg p \wedge q \wedge r)$
 D. $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$

Answer key

11.6.9 Propositional Logic: UGC NET CSE | June 2019 | Part 2 | Question: 6



Which of the following is principal conjunctive normal form for $[(p \vee q) \wedge \neg p \rightarrow \neg q]$?

- A. $p \vee \neg q$
 B. $p \vee q$
 C. $\neg p \vee q$
 D. $\neg p \vee \neg q$

Answer key

11.6.10 Propositional Logic: UGC NET CSE | June 2019 | Part 2 | Question: 8



Match List-I with List-II:

	List-I		List-II
(a)	$p \rightarrow q$	(i)	$\neg(q \rightarrow \neg p)$
(b)	$p \vee q$	(ii)	$p \wedge \neg q$
(c)	$p \wedge q$	(iii)	$\neg p \rightarrow q$
(d)	$\neg(p \rightarrow q)$	(iv)	$\neg p \vee q$

Choose the correct option from those given below:

- A. (a) – (ii); (b) – (iii); (c) – (i); (d) – (iv)
 B. (a) – (ii); (b) – (i); (c) – (iii); (d) – (iv)
 C. (a) – (iv); (b) – (i); (c) – (iii); (d) – (ii)
 D. (a) – (iv); (b) – (iii); (c) – (i); (d) – (ii)

ugcnetcse-june2019-paper2 propositional-logic

Answer key 

11.6.11 Propositional Logic: UGC NET CSE | Junet 2015 | Part 2 | Question: 10



Consider a proposition given as:

$x \geq 6$, if $x^2 \geq 25$ and its proof as:

If $x \geq 6$, then $x^2 = x \cdot x \geq 6 \cdot 6 = 36 \geq 25$

Which of the following is correct with respect to the given proposition and its proof?

- i. The proof shows the converse
 ii. The proof starts by assuming what is to be shown
 iii. The proof is correct and there is nothing wrong

- A. *i* only B. *iii* only C. *i* and *ii* D. *ii* only

ugcnetcse-june2015-paper2 discrete-mathematics propositional-logic

Answer key 

11.6.12 Propositional Logic: UGC NET CSE | Junet 2015 | Part 2 | Question: 9



Match the following :

List-I

- (a) $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
 (b) $[(p \wedge q) \rightarrow r] \Leftrightarrow [p \rightarrow (q \rightarrow r)]$
 (c) $(p \rightarrow q) \Leftrightarrow [(p \wedge \neg q) \rightarrow o]$
 (d) $(p \leftrightarrow q) \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$

List-II

- (i) Contrapositive
 (ii) Exportation law
 (iii) Reduction as absurdum
 (iv) Equivalence

Codes :

- A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) B. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 C. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) D. (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

ugcnetcse-june2015-paper2 mathematical-logic propositional-logic

Answer key 

11.6.13 Propositional Logic: UGC NET CSE | October 2020 | Part 2 | Question: 61



Consider the statement below.

A person who is radical (R) is electable (E) if he/she is conservative (C), but otherwise not electable.

Few probable logical assertions of the above sentence are given below.

- i. $(R \wedge E) \Leftrightarrow C$
 ii. $R \rightarrow (E \leftrightarrow C)$
 iii. $R \Rightarrow ((C \Rightarrow E) \vee \neg E)$
 iv. $(\neg R \vee \neg E \vee C) \wedge (\neg R \vee \neg C \vee E)$

Which of the above logical assertions are true?

Choose the correct answer from the options given below:

- A. (*ii*) only B. (*iii*) only C. (*i*) and (*iii*) only D. (*ii*) and (*iv*) only

ugcnetcse-oct2020-paper2 discrete-mathematics propositional-logic

Answer key 

11.6.14 Propositional Logic: UGC NET CSE | October 2022 | Part 1 | Question: 85



Consider α, β, γ as logical variables. Identify which of the following represents correct logical equivalence :

- (A) $(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$
- (B) $(\alpha \vee \beta) \equiv \neg \alpha \vee \beta$
- (C) $(\alpha \Rightarrow \beta) \equiv (\neg \beta \Rightarrow \neg \alpha)$
- (D) $(\neg(\alpha \vee \beta)) \equiv (\neg \alpha \Rightarrow \neg \beta)$

Choose the correct answer from the options given below :

- A. (A) and (D) only
- B. (B) and (C) only,
- C. (A) and (C) only
- D. (B) and (D) only

ugcnetcse-oct2022-paper1 propositional-logic

11.6.15 Propositional Logic: UGC NET CSE | September 2013 | Part 2 | Question: 24



The quantification $\exists! x P(x)$ denotes the proposition “There exists a unique x such that $P(x)$ is true”, express he quantification using universal and existential quantifications and logical operators?

- A. $\exists x P(x) \vee \forall x \forall y ((P(x) \vee P(y)) \rightarrow x = y)$
- B. $\forall x P(x) \wedge \forall x \forall y ((P(x) \vee P(y)) \rightarrow x = y)$
- C. $\exists x P(x) \wedge \forall x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$
- D. $\exists x P(x) \wedge \forall x \forall y ((P(x) \vee P(y)) \rightarrow x = y)$

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Answer key

11.7 Relations (1)

11.7.1 Relations: UGC NET CSE | October 2020 | Part 2 | Question: 39



Consider the following properties:

- i. Reflexive
- ii. Antisymmetric
- iii. Symmetric

Let $A = \{a, b, c, d, e, f, g\}$ and $R = \{(a, a), (b, b), (c, d), (c, g), (d, g), (e, e), (f, f), (g, g)\}$ be a relation on A . Which of the following property (properties) is (are) satisfied by the relation R ?

- A. Only i
- B. Only iii
- C. Both i and ii
- D. ii and not i

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Answer key

Answer Keys

11.0.1	C	11.0.2	D	11.0.3	D	11.0.4	B	11.0.5	B
11.0.6	A	11.0.7	C	11.0.8	B	11.0.9	A	11.0.10	A
11.0.11	X	11.0.12	Q-Q	11.0.13	A;B	11.0.14	D	11.0.15	A
11.0.16	A	11.0.17	C	11.0.18	Q-Q	11.0.19	Q-Q	11.1.1	D
11.2.1	B	11.2.2	A	11.2.3	C	11.2.4	A	11.2.5	A
11.3.1	Q-Q	11.4.1	D	11.5.1	A	11.6.1	Q-Q	11.6.2	Q-Q
11.6.3	C	11.6.4	B;C	11.6.5	A;B;C;D	11.6.6	B	11.6.7	Q-Q
11.6.8	D	11.6.9	A	11.6.10	D	11.6.11	C	11.6.12	A
11.6.13	D	11.6.14	Q-Q	11.6.15	C	11.7.1	D		